

Can 50 Logical Qubits Beat Classical Limits for the Hubbard Model?

Guidance for the EIGEN-Q proposal on the Magne system

Technical assessment prepared for M. Hjorth-Jensen & the EIGEN-Q team

June 2026

Neutral-atom Magne · 50 logical / $\sim 1,225$ physical qubits · online $\approx 2026/27$

Bottom line up front

The honest one-sentence answer

For *generic ground-state energies* of the 2D Hubbard model, **50 logical qubits will not beat classical methods** — but for a *narrow, genuinely hard niche* (frustrated + doped lattices and real-time dynamics) it can produce results that classical methods cannot reach reliably.

- 50 logical qubits $\Rightarrow$ **~ 24 – 25 lattice sites** (2 qubits/site).
- Your own classical reach is already **larger**: ED to 24 sites, DMRG to 48 sites.
- So “advantage” cannot come from *size*. It must come from the **regime**: where the sign problem and entanglement defeat QMC and DMRG.
- The **binding constraint** on Magne will be circuit *depth* (logical error rate), not qubit count.

Recommendation: frame Hubbard work as a credible *stepping-stone demonstration* on classically-hard regimes, not as ground-state quantum advantage.

How many sites does 50 logical qubits actually buy?

The Hubbard site carries two spin-orbitals ($\uparrow, \downarrow$). Under Jordan–Wigner / Bravyi–Kitaev each spin-orbital = one qubit:

$$L \text{ sites} \longrightarrow 2L \text{ qubits.}$$

- 50 logical qubits $\Rightarrow L \approx 25$ sites (e.g. a 5×5 cluster).
- Rodeo needs 1–3 ancillas $\Rightarrow \sim \mathbf{23\text{--}24}$ **physical sites**.
- $t\text{--}J$ / infinite- U : local space is $\{0, \uparrow, \downarrow\}$ (3 states). A compact (qutrit-like) encoding can recover a few extra sites, $\sim \mathbf{30}$ **sites at**

Sanity check on the text

The application mentions “Hubbard clusters with more than 30 sites” at 20–40 qubits. With 2 qubits/site that needs **60+ qubits** — inconsistent with 50 logical. Either state the compact encoding explicitly, or revise to ≤ 25 sites.

Where the classical frontier really is

Method	Reach for 2D Hubbard	What breaks it
Exact diagonalisation	~ 20 – 22 sites (routine); 32 – 36 sites (extreme HPC)	memory: Hilbert space $\sim 4^L$
DMRG / MPS	1D: hundreds of sites; 2D cyl. width $\lesssim 6$ – 8 ; your 48	2D area law: bond dim. grows exponentially with width
AFQMC / DQMC	hundreds of sites <i>only</i> where no sign problem	sign problem (doped & frustrated)

Consequence

A ~ 25 -site quantum result sits *at or below* the existing classical reach for **static ground-state energy**. The cross-validated Simons-collaboration benchmarks already pin down 2D square-lattice Hubbard energies far beyond this size.

The real classical wall is not size — it is sign & entanglement

Where classical methods genuinely fail:

- **Frustration** (checkerboard, pyrochlore) $\Rightarrow$ QMC sign problem *even near half-filling*.
- **Doping / finite hole density** $\Rightarrow$ sign problem for QMC.
- **Spin-liquid regime** $\Rightarrow$ high entanglement $\Rightarrow$ DMRG bond dimension explodes.
- **Dynamics / spectral functions / excited states** $\Rightarrow$ time-entanglement growth (DMRG) and analytic continuation (QMC)

This is precisely hubbard.txt

Your notes already identify the wall:

- “DMRG ... rather hopeless to go beyond two holes... entanglement ... major obstacle... quantum spin-liquid regime.”
- Analytical RVB proof holds only for **1–2 holes**; **3–4 holes** (finite density) is open.
- Finite U phase diagram is open.

So — is ~ 25 sites “beyond classical”? A layered answer

Target	Verdict at 50 logical qubits
2D square Hubbard, GS energy, half / weak doping	No advantage. Classical does this far larger.
Frustrated lattice, 1–2 holes	Useful as <i>validation</i> (you have ED/DMRG/proof here).
Frustrated lattice, 3–4 holes / finite density	Genuinely hard classically — sign problem + entanglement. A reliable 24–30-site result <i>is</i> new.
Real-time dynamics, spectral functions, excited states	Structural quantum advantage candidate even at modest size.

The defensible advantage claim lives in the bottom two rows — exactly the open problems in your notes. Finite-size effects on a 5×5 frustrated cluster are real, so pair every run with a careful finite-size analysis.

The binding constraint is depth, not width

- Magne packs $\sim 1,225$ physical into 50 logical $\Rightarrow \sim 24$ **physical per logical** $\Rightarrow$ a *low* code distance ($d \approx 3-5$).
- Low distance $\Rightarrow$ a logical error floor that caps the number of logical operations you can chain before errors accumulate.
- Your pipeline is **depth-hungry**: adiabatic refinement ($\propto T^*$ Trotter steps) + M rounds of controlled- e^{-iHt_j} . For ~ 25 sites this is easily 10^3-10^4 logical two-qubit gates.

Implication

The first question is not “do we have enough qubits?” (you do, for ~ 25 sites) but “**can the logical error rate sustain the circuit depth?**”. This must be settled by *classical emulation with a realistic Magne noise model* before hardware time is committed.

What the published resource estimates say

- Ground-state advantage for the 2D Fermi–Hubbard model is estimated at $\sim 10^5$ **physical qubits** (surface code, $p = 10^{-3}$). [npj Quantum Inf. **10**, 2024; arXiv:2210.14109]
- Even at $p = 10^{-4}$, beating classical for the 8×8 Hubbard model needs $\sim 6.5 \times 10^4$ **physical qubits**. [arXiv:2408.14929]
- Finding the Hubbard ground state is **QMA-hard**; frustrated / triangular / three-band variants are *expected* to require quantum resources. [perf.-model study, ACM 2023]

Reading this for Magne

Magne's $\sim 1,225$ physical qubits are **1–2 orders of magnitude** below the threshold these studies set for *ground-state* 2D-Hubbard advantage. That is not a flaw — it tells you Magne's value is as a **Level-2 demonstration and algorithm-validation platform**, and as the on-ramp to the machines that will cross the line.

Where the genuine opportunity is for *this* team

- 1 Frustrated t - J with 3–4 holes.** Sign problem (QMC) + entanglement (DMRG) make this classically unreliable; 24–30 sites with refinement+rodeo is a real target and directly extends your RVB-stability question to finite density.
- 2 Dynamics & excited states of the RVB spin liquid.** Spectral functions $S(\mathbf{q}, \omega)$, dynamical structure factors, and low-lying excitations — a structural-advantage regime, and it answers your open “which observables?” question.
- 3 Finite- U phase boundary.** Map RVB stability vs. U/t where the $J \neq 0$ results already hint the answer is “yes for $U \gg t$ ”.
- 4 Method head-to-head.** Demonstrate refinement+rodeo is *more efficient than DMRG* on systems up to 48 sites — exactly the claim your notes flag as valuable even if Magne cannot exceed 48 sites.

Concrete recommendations for the proposal

Framing

- Drop any implicit “GS-energy quantum advantage” claim for 2D Hubbard; reviewers know the resource estimates.
- Position Hubbard work as **(a)** hard-regime demonstration and **(b)** algorithm validation / stepping-stone.
- State the advantage targets explicitly: *sign-problem regimes* and *dynamics*.

Technical

- Fix the “ > 30 sites at 20–40 qubits” line, or specify a compact t - J encoding.

De-risking (do first)

- Build a **logical depth budget**: Trotter steps $\times$ rodeo rounds $\times$ gate count, folded against Magne’s projected logical error rate.
- Classically emulate the full pipeline *with realistic noise* on 4–16 sites before requesting hardware.

Benchmark ladder

- 4–6 sites: validate vs. ED.
- 16 sites, 1–2 holes: validate vs. your ED/DMRG/proof.
- 24–30 sites, 3–4 holes: the *new science*.

Closing summary

Can you move beyond classical limits with 50 logical qubits?

Not for Hubbard ground-state energies in general — 25 sites is at or below classical reach, and depth (not qubits) is the limiter. **Yes, plausibly, in a focused niche:** frustrated lattices at finite hole density and real-time/dynamical observables, where the sign problem and entanglement defeat classical methods even at modest size.

- Treat Magne as a **Level-2 demonstrator** on classically-hard regimes — scientifically valuable and fundable on its own terms.
- Validate the depth budget under noise *first*; let that, not qubit count, gate the targets.
- Aim the science where your team is unique: **RVB at finite doping** and its dynamics.

Key references used

- Magne / QuNorth specifications — EIFO & Novo Nordisk Foundation announcement (2025); Quantum Computing Report (2026): 50 logical / 1,225 physical neutral-atom qubits.
- “Hunting for quantum-classical crossover in condensed matter problems,” npj Quantum Inf. **10** (2024) / arXiv:2210.14109 ($\sim 10^5$ physical qubits for 2D Hubbard GS).
- “Compilation of Trotter-based time evolution . . . ,” arXiv:2408.14929 ($\sim 6.5 \times 10^4$ physical qubits for 8×8 at $p = 10^{-4}$).
- “A performance model for . . . practical quantum advantage,” ACM (2023) (Hubbard GS is QMA-hard; frustrated variants expected to need quantum).
- Tohyama *et al.*, Phys. Rev. B **72**, 045113 (2005) (ED to 20 sites in 2D); Tianhe-2 distributed Lanczos to 36 sites.
- Internal: `application.tex` (EIGEN-Q) and `hubbard.txt` (RVB / frustrated t - J notes).